

ECE 313: Problem Set 6: Problems and Solutions

Due: Friday, March 14 at 7:00:00 p.m.

Reading: *ECE 313 Course Notes*, Sections 3.2 - 3.5

Note on reading: For most sections of the course notes there are short answer questions at the end of the chapter. We recommend that after reading each section you try answering the short answer questions. Do not hand these in; answers to the short answer questions are provided in the appendix of the notes.

Note on turning in homework: Homework is assigned on a weekly basis on Fridays, and is due by 7 p.m. on the following Friday. **Please write down your work and derivations. An answer without justification as of how it is found will not be accepted.** You must upload handwritten homework to Gradescope. Alternatively, you can typeset the homework in LaTeX. However, no additional credit will be awarded to typeset submissions. No late homework will be accepted.

Please write on the top right corner of the first page:

NAME

NETID

SECTION

PROBLEM SET #

Page numbers are encouraged but not required. Five points will be deducted for improper headings. Please assign your uploaded pages to their respective question numbers while submitting your homework on Gradescope. **5 points will be deducted for incorrectly assigned page numbers.**

1. **[Probability Density Functions]**

The lifetime of a certain kind of radio tube is a random variable X with pdf

$$f(x) = \frac{C}{x^2}, \text{ for } x > 100,$$

and zero otherwise.

- (a) Find the constant C .

Solution: We need to find C such that $\int_{100}^{\infty} \frac{C}{x^2} dx$

$$\begin{aligned} 1 &= \int_{100}^{\infty} \frac{C}{x^2} dx \\ &= \left(-\frac{C}{x} \right) \Big|_{100}^{\infty} \\ &= \left(0 + \frac{C}{100} \right) \\ &\implies C = 100 \end{aligned}$$

- (b) Assuming that tubes break down independently, find the probability that at least 3 out of 5 tubes will break down within the first 150 hours of operation.

Solution: The probability of a single radio tube failing within the first 105 hours is $P(X \leq 150)$. Solving this, we get

$$\begin{aligned}\int_{100}^{150} \frac{100}{x^2} dx &= \left(-\frac{100}{x}\right) \Big|_{100}^{150} \\ &= \frac{1}{3}\end{aligned}$$

If we have 5 radio tubes, the probability of at least three of them failing can be understood as $P(Y \geq 3)$, where $Y \sim \text{Bin}(5, \frac{1}{3})$.

Then,

$$\begin{aligned}P(Y \geq 3) &= \binom{5}{3} \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^2 + \binom{5}{4} \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^1 + \binom{5}{5} \left(\frac{1}{3}\right)^5 \left(\frac{2}{3}\right)^0 \\ &= \frac{17}{81} \approx 0.2099\end{aligned}$$

(c) What can you say about the expectation and variance of X ?

Solution:

$$\begin{aligned}E[X] &= \int_{100}^{\infty} \frac{100}{x} dx \\ &= (100 \ln(x)) \Big|_{100}^{\infty} \\ &\rightarrow \infty\end{aligned}$$

The second moment $E[X^2]$ is also not finite, so the variance is also not finite.

2. [Exponential Random Variables]

The amount of time, in hours, of a computer working without breaking down is a random variable with pdf

$$f(x) = \lambda \exp\left(-\frac{x}{100}\right), \text{ for } x > 0,$$

and zero otherwise.

(a) Find λ .

Solution:

$$\begin{aligned}1 &= \int_0^{\infty} \lambda \exp\left(-\frac{x}{100}\right) dx \\ &= \lambda \left(-100 \exp\left(-\frac{x}{100}\right)\right) \Big|_0^{\infty} \\ &= \lambda(0 + 100) \\ \implies \lambda &= \frac{1}{100}\end{aligned}$$

(b) What is the probability that the computer will function between 50 and 150 hours before breaking down?

Solution: In other words, what is $P(50 < X < 150)$? We can solve this question by first finding the closed-form CDF of X .

$$\begin{aligned}
F_X(c) &= \frac{1}{100} \int_0^c \exp\left(-\frac{x}{100}\right) dx \\
&= \frac{1}{100} \left(-100 \exp\left(-\frac{x}{100}\right)\right)_0^c \\
&= -1 \left(\exp\left(-\frac{c}{100}\right) - 1\right) \\
&= 1 - \exp\left(-\frac{c}{100}\right)
\end{aligned}$$

Since X is a continuous type random variable, $P(X = 150) = 0$ and thus $F_X(150) = P(X < 150) + P(X = 150) = P(X < 150)$. Then,

$$\begin{aligned}
P(50 < X < 150) &= P(X < 150) - P(X \leq 50) \\
&= F_X(150) - F_X(50) \\
&= 1 - \exp\left(-\frac{150}{100}\right) - \left(1 - \exp\left(-\frac{50}{100}\right)\right) \\
&= \exp\left(-\frac{1}{2}\right) - \exp\left(-\frac{3}{2}\right)
\end{aligned}$$

- (c) What is the probability that the computer will function less than 100 hours?

Solution: Using the CDF we found before,

$$\begin{aligned}
P(X < 100) &= F_X(100) \\
&= 1 - \exp\left(-\frac{100}{100}\right) \\
&= 1 - \frac{1}{e}
\end{aligned}$$

3. [Uniform Continuous Random Variables]

A stick of length 2 is split at a point U that is uniformly distributed over $(0, 2)$.

- (a) Determine the expected length of the part of the stick that contains a given point p .

Solution: Let U be the RV that stands for the length from 0 to the cut. U has density f_U uniform in $[0, 2]$.

We denote by $\ell(U)$ the RV (which is a function of U) which stands for the length of the piece that contains point p . If $U = u$ then

$$\ell(u) = \begin{cases} 2 - u, & u \in [0, p) \\ u, & u \in [p, 2]. \end{cases}$$

Therefore, using LOTUS:

$$\begin{aligned}
\mathbb{E}[\ell(U)] &= \int_0^2 f_U(u) \ell(u) du = \int_0^p \frac{1}{2} \cdot (2 - u) du + \int_p^2 \frac{1}{2} \cdot u du \\
&= \frac{1}{2} \left[2u - \frac{u^2}{2} \right]_0^p + \frac{1}{2} \left[\frac{u^2}{2} \right]_p^2 \\
&= \frac{1}{2} \left[2p - \frac{p^2}{2} \right] + \frac{1}{2} \left[2 - \frac{p^2}{2} \right] \\
&= 1 + p - \frac{p^2}{2}.
\end{aligned}$$

- (b) Which choice of p maximizes the expected length of the piece described in the previous part?

Solution: This function is convex-down, so it has a global maximum. We can take the derivative and set it to 0.

The derivative of this function is $1 - p$, so setting $p = 1$ maximizes the expected length of the stick to $\frac{3}{2}$.

4. [Departure time]

Suppose that if you are s minutes early for an appointment, you incur a cost of cs , $c > 0$, and if you are s minutes late, you incur a cost ks , $k > 0$. Also, suppose that the travel time from your house to the location of your appointment is a continuous random variable X with pdf $f(x)$ and CDF $F_X(x)$. Determine the time at which you should depart if you want to minimize your expected appointment cost. (Hint: You may find the Leibniz integral rule useful for minimizing your expected appointment cost).

Solution: Assume you leave your house at $T = T_s$, and let $T = T_p$ be the appointment time. Then, your arrival time T_a can be expressed as

$$T_s + X = T_a$$

Let $C(T_a)$ be our cost function. Then,

$$C(T_a) = \begin{cases} c(T_p - T_a) & T_a < T_p \\ 0 & T_a = T_p \\ k(T_a - T_p) & T_a > T_p \end{cases}$$

Formulating this in terms of X gives

$$\tilde{C}(X) = \begin{cases} c(T_p - T_s - X) & X < T_p - T_s \\ 0 & X = T_p - T_s \\ k(X + T_s - T_p) & X > T_p - T_s \end{cases}$$

Let $t = T_p - T_s$ i.e. the number of minutes early you leave. Then,

$$\tilde{C}(X) = \begin{cases} c(t - X) & X < t \\ 0 & X = t \\ k(X - t) & X > t \end{cases}$$

We first solve for $E[\tilde{C}(X)]$.

$$\begin{aligned} E[\tilde{C}(X)] &= \int_0^\infty \tilde{C}(x)f(x)dx \\ &= \int_0^t c(t-x)f(x)dx + \int_t^\infty k(x-t)f(x)dx \\ &= ct \int_0^t f(x)dx - c \int_0^t xf(x)dx + k \int_t^\infty xf(x)dx - kt \int_t^\infty f(x)dx \\ &= ctF_X(t) - c \int_0^t xf(x)dx + k \int_t^\infty xf(x)dx - kt[1 - F_X(t)] \end{aligned}$$

Using the Leibniz integral rule on the middle two integrals,

$$\begin{aligned}\frac{d}{dt}E[\tilde{C}(X)] &= cF_X(t) + ctf(t) - ctf(t) - ktf(t) - k + kF_X(t) + ktf(t) \\ &= cF_X(t) - k + kF_X(t)\end{aligned}$$

This equals zero when $F_X(t) = \frac{k}{k+c}$ and it is easy to verify that it corresponds to the global minima. As a result, we should depart t^* minutes before the appointment time where $F_X(t^*) = \frac{k}{k+c}$.

5. [Poisson Processes]

Let $\{N(t), t \in [0, \infty)\}$ be a Poisson Process with rate $\lambda = \frac{1}{2}$.

- (a) Find the probability of no arrivals in the intervals $[0, 1]$, $[1, 2]$ and $[2, 3]$. Compare these probabilities with each other and with the probability of no arrivals in $[0, 3]$.

Solution: For $[0, 1]$, we have rate $\lambda_{1-0} = \lambda_1 = 1 \cdot \lambda = \frac{1}{2}$.

$$\begin{aligned}P(\text{no arrivals in } [0, 1]) &= P(N_1 = 0) \\ &= \frac{e^{-\frac{1}{2}}(\frac{1}{2})^0}{0!} \\ &= e^{-\frac{1}{2}}\end{aligned}$$

Since the increments $[0, 1]$, $[1, 2]$, $[2, 3]$ have the same duration, i.e. their rates are all λ_1 , they also have the same distribution. However, $[0, 3]$ is different, and has rate $\lambda_{3-0} = \lambda_3 = 3 \cdot \lambda = \frac{3}{2}$.

$$\begin{aligned}P(\text{no arrivals in } [0, 3]) &= P(N_3 = 0) \\ &= \frac{e^{-\frac{3}{2}}(\frac{3}{2})^0}{0!} \\ &= e^{-\frac{3}{2}}\end{aligned}$$

This is lower than the probability of there being no arrivals in any unit interval.

- (b) Find the probability that there are exactly 1, 2, and 3 arrivals in the intervals $[0, 1]$, $[2, 4]$ and $[\frac{1}{2}, 3]$, respectively.

Solution: Due to ambiguity, we accepted two different answers.

To find $P(N_1 = 1)$, $P(N_4 - N_2 = 2)$, and $P(N_3 - N_{\frac{1}{2}} = 3)$, we do the following: The rates are $\frac{1}{2}$, 1, and $\frac{5}{4}$, respectively.

$$\begin{aligned}P(1 \text{ arrival in } [0, 1]) &= P(N_1 = 1) \\ &= \frac{e^{-\frac{1}{2}}(\frac{1}{2})^1}{1!} \\ &= \frac{1}{2}e^{-\frac{1}{2}}\end{aligned}$$

$$\begin{aligned}P(2 \text{ arrival in } [2, 4]) &= P(N_4 - N_2 = 2) \\ &= \frac{e^{-1}(1)^2}{2!} \\ &= \frac{1}{2}e^{-1}\end{aligned}$$

$$\begin{aligned}
P(3 \text{ arrival in } [\frac{1}{2}, 3]) &= P(N_3 - N_{\frac{1}{2}} = 3) \\
&= \frac{e^{-\frac{5}{4}} (\frac{5}{4})^3}{3!} \\
&= \frac{125}{384} e^{-\frac{5}{4}}
\end{aligned}$$

To find $P(N_1 = 1, N_4 - N_2 = 2, N_3 - N_{\frac{1}{2}} = 3)$, we do the following:

The intervals that need to be considered are $N_{\frac{1}{2}} - N_0$, $N_1 - N_{\frac{1}{2}}$, $N_2 - N_1$, $N_3 - N_2$, and $N_4 - N_3$. There are 6 total cases of independent intervals that satisfy the original event.

$$N_{\frac{1}{2}} - N_0 = 0, N_1 - N_{\frac{1}{2}} = 1, N_2 - N_1 = 0, N_3 - N_2 = 2, N_4 - N_3 = 0 \quad (1)$$

$$N_{\frac{1}{2}} - N_0 = 0, N_1 - N_{\frac{1}{2}} = 1, N_2 - N_1 = 1, N_3 - N_2 = 1, N_4 - N_3 = 1 \quad (2)$$

$$N_{\frac{1}{2}} - N_0 = 0, N_1 - N_{\frac{1}{2}} = 1, N_2 - N_1 = 2, N_3 - N_2 = 0, N_4 - N_3 = 2 \quad (3)$$

$$N_{\frac{1}{2}} - N_0 = 1, N_1 - N_{\frac{1}{2}} = 0, N_2 - N_1 = 1, N_3 - N_2 = 2, N_4 - N_3 = 0 \quad (4)$$

$$N_{\frac{1}{2}} - N_0 = 1, N_1 - N_{\frac{1}{2}} = 0, N_2 - N_1 = 2, N_3 - N_2 = 1, N_4 - N_3 = 1 \quad (5)$$

$$N_{\frac{1}{2}} - N_0 = 1, N_1 - N_{\frac{1}{2}} = 0, N_2 - N_1 = 3, N_3 - N_2 = 0, N_4 - N_3 = 2 \quad (6)$$

Then,

$$\begin{aligned}
P(N_1 = 1, N_4 - N_2 = 2, N_3 - N_{\frac{1}{2}} = 3) &= P((1) \cup (2) \cup (3) \cup (4) \cup (5) \cup (6)) \\
&= P((1)) + P((2)) + P((3)) + P((4))P((5)) + P((6)) \\
&= e^{-4}\lambda \left(\frac{\lambda^3}{4} + \frac{\lambda^4}{2} + \frac{\lambda^5}{8} + \frac{\lambda^4}{4} + \frac{\lambda^5}{4} + \frac{\lambda^6}{24} \right) \\
&= e^{-2} \left(\frac{1}{24} + \frac{1}{32} + \frac{1}{256} + \frac{1}{64} + \frac{1}{128} + \frac{1}{1536} \right) \\
&= \frac{125}{1536} e^{-2}
\end{aligned}$$

- (c) Let X_1 be the first arrival time in the Poisson Process. Show that, given $N_t = 1$, X_1 is uniformly distributed in $(0, t]$.

Solution: Let c be some time between 0 and t .

$$\begin{aligned}
P(X_1 \leq c | N_t = 1) &= \frac{P(X_1 \leq c \cap N_t = 1)}{P(N_t = 1)} \\
&= \frac{P(N_c = 1 \cap N_t - N_c = 0)}{P(N_t = 1)} \\
&= \frac{P(N_c = 1)P(N_t - N_c = 0)}{P(N_t = 1)} \\
&= \frac{\frac{e^{-\frac{1}{2}c} (\frac{1}{2}c)^1}{1!} \frac{e^{-\frac{1}{2}(t-c)} (\frac{1}{2}(t-c))^0}{0!}}{\frac{e^{-\frac{1}{2}t} (\frac{1}{2}t)^1}{1!}} \\
&= \frac{c}{t}
\end{aligned}$$

We see that this conditional CDF is the same as the CDF of a uniform random variable on $[0, t]$.